

iQOO CITY BATTLES 2026 • HYDERABAD

TaskLens

Capture once. Learn anywhere.
Work without the internet.

PRESENTED BY TEAM NEXURA

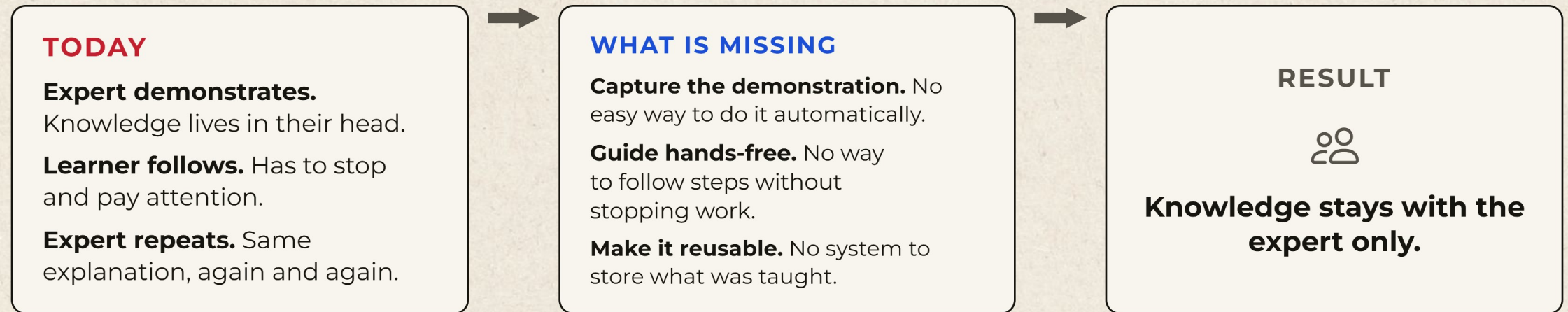
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Problem

- Most practical work knowledge lives only in people's heads: **the right order of steps, the important checks, the small corrections** that are hard to write down.
- Manuals take too long to write, and videos are hard to watch when **your hands are busy**. So people rely on **live demonstrations**, repeated for every new person.
- **886M Indians are online and 98% consume content in Indian languages**. Yet guides stay English-first, and voice assistants struggle in noisy, hands-busy rooms.

The Core Gap



Solution

TaskLens is one app with two modes. The person who knows the job does it once and talks through it. Everyone after that is taught by the phone: hands-free, in that person's own voice, with no internet and everything on the device.

SHOW

once, by the person who knows

- The expert **does the job once and explains it aloud** while the phone watches and listens.
- Pauses and words like “phir” or “then” **cut the take into steps**, one photo each.
- A 60–90 second take becomes a guide; the expert reviews it and **signs it off**.

record → transcribe → cut → photo → verify

GUIDE

after that, for everyone else

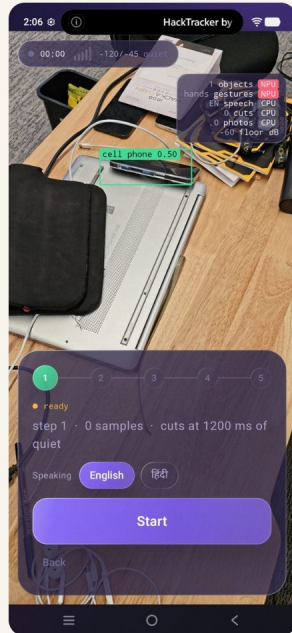
- The next person works with the phone nearby. It reads **how it is held and how loud the room is**.
- It picks **TAP, TALK or HANDS** by itself, and always shows why.
- Each step plays in **the expert's own voice**; palm and fist move between steps.

watch → decide → play the step → check the scene

How it works

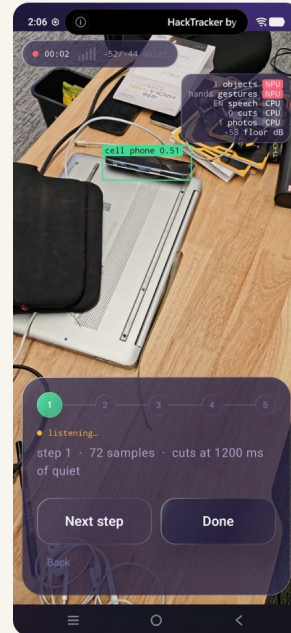
SHOW

Record once, by the expert. Real screens from the app.



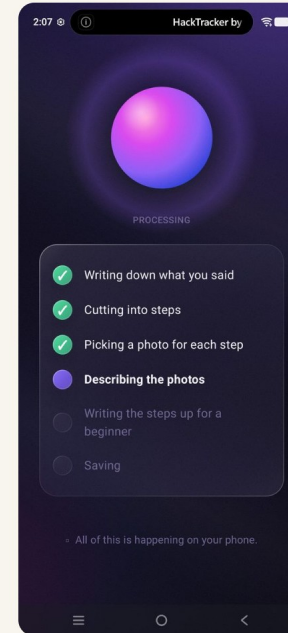
1. Ready

Pick Hindi or English. A HUD shows which chip runs each model.



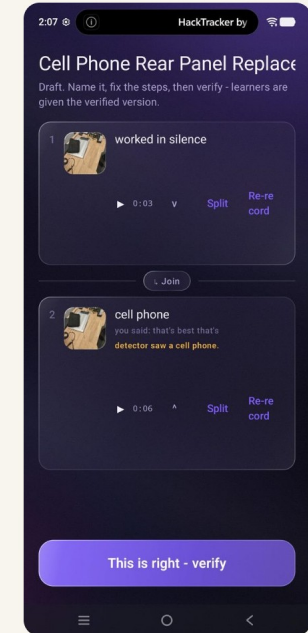
2. Talk

Cuts at 1200 ms of quiet. The noise floor is tracked live.



3. Process

Six stages, all on the phone. Nothing is uploaded.



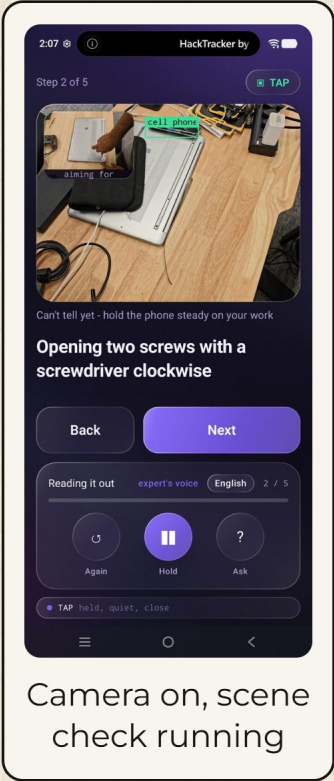
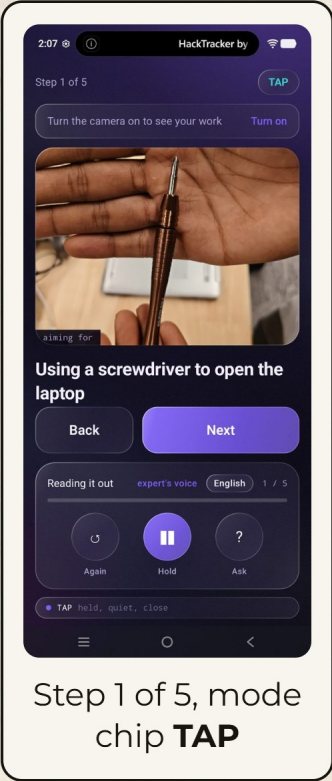
4. Review

Split, join, re-record. Learners see only what the expert signs off.

How it works

GUIDE

Anyone runs it, hands full. The phone reads the room.



Mode	When	What the phone does
EASY	Chosen in settings	Bigger text, slower, no guessing
HANDS	Room is loud, or speech unclear	Camera on: palm and fist move ± 1 step
TALK	Phone flat on a counter	Big text, voice reads the step
TAP	Held, quiet, close	Normal cards and buttons

No AI in this path

Plain rules over four booleans, first match wins, decided in under 10 ms. A 400 ms dwell stops flicker. Every switch shows its reason, like **HANDS** ← **room is loud (-24.3 dBFS)**.

The adaptive gate

A fixed loudness threshold cannot cut steps in a real room. **A ceiling fan floors a kitchen near -20 dB**, so a gate nailed at -38 dB hears endless speech, finds no pause, and returns one enormous step.

-6 dB

Loud speech peaks near here

-11 dB

Tracked gate: floor + 9 dB margin. Speech must clear it.

-20 dB

Ceiling-fan floor: the room is never quieter

-38 dB

A fixed gate: the fan counts as speech, no pause is ever found

So the gate tracks the floor

- **Falls fast:** from 0 to a -20 dB fan in about half a second.
- **Rises slowly,** and freezes while someone speaks, so a long sentence cannot drag it up.
- A dead mic or a NaN is treated as -120 dB, never a crash.

Held by unit tests for the fan-floored room, the clamp, the silent mic and the long utterance.

Six models, one phone

Model	Job	If it is absent
Vosk	Hindi and English speech, with word timings	Pause detector stands alone
Gemma 3 1B int4	The coach: rewrites steps, titles the guide, answers questions	Steps stay in the expert's words
Tool detector	Screwdriver and five screw-head types (fine-tuned)	No tool boxes
COCO detector	Laptop, keyboard, mouse, person	The room goes unnamed
Gesture recognizer	Palm and fist move ± 1 step	Buttons still work
Scene check	Does this bench match the guide's photo? No model file	Never absent

Coach, measured once

Loads on the GPU in 15.9 s, answers a question in 6.9 s. Memory use is still unmeasured.

Tool detector

AP@0.50 of 0.635, weak on small screw heads (AP small 0.157). Runs on the NPU.

Never faked

A missing model is never replaced by a stand-in. The app drops a tier and says so.

The honest part

Four hard constraints

- **No INTERNET permission.** The manifest declares only camera and microphone. The app-info screen proves it in four seconds.
- **The step cutter is a pause detector plus a word list.** No ML in that path.
- **The mode engine contains no AI.**
- **Warnings advise, they never block.** No disabled button anywhere.

What the phone will not guess

- **Gloves, driving, age, and whether the work is correct.** These are settings, never guesses.
- **How far away the user is.** Not measured yet, so it stays off.
- **Marathi recording.** No offline speech model exists for it, and a button that returns nothing is the failure we refuse.

Provenance. The coach is the only model allowed to say something the expert did not, so every instruction carries a label:

EXPERT

VISUAL

GENERAL

UNKNOWN

Phone-first

The iQOO 15 is the camera, the writer, the teacher and the checker. **Four jobs, one phone.** Take the phone away and there is nothing left to show.

Office Kit	We will use all four functions across the 30 hours. File transfer moves the model files and every policy.json we tune. Screen mirror puts the phone on a projector. Shared clipboard carries run logs. Remote control fixes step wording from the laptop keyboard.					
Technical depth	AI only where language and vision need it, plain code everywhere else. The step cutter is a pause detector and a word list. Every number lives in one policy.json that reloads live. We show which chip runs each model, and the things we refuse to guess.					
End product quality	Two judges, one phone, airplane mode, 2.5 minutes. One records a job; the other, who has never seen it, is taught by the phone. The guide they walk away with did not exist when they sat down.					
<table><tr><td>MAKING A GUIDE How long a 60–90 second take needs, stage by stage.</td><td>SPEED How fast modes switch and a palm is recognised.</td><td>WORKING OFFLINE Memory and chip use with every model loaded.</td></tr></table>				MAKING A GUIDE How long a 60–90 second take needs, stage by stage.	SPEED How fast modes switch and a palm is recognised.	WORKING OFFLINE Memory and chip use with every model loaded.
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The 30-hour sprint

10.5 of the 30 hours are Red Light, with the laptop closed as a build machine. Rules, word lists and guides are text files, so Red Light is spent building, not waiting.

When	Light	What we build
Sat 11:00–13:00	GREEN	Rebuild on the loaner phone and push every model file over Office Kit.
Sat 13:00–15:30	RED	Show Mode end to end: recording, live transcript, step cutter, review screen.
Sat 15:30–16:30	GREEN	Breather for anything that needs a laptop, including measuring the coach's memory use.
Sat 16:30–19:00	RED	Tune policy.json: word lists, gate, dwell. A text file, so no rebuild.
Sat 22:00–Sun 01:00	RED	Guide playback: voice clips, the mode reason bar, photo checking.
Sun 01:00–06:30	GREEN	Heavy integration, Office Kit testing and a one-hour heat test.
Sun 06:30–09:00	RED	Record five real guides. Practise in airplane mode. Break it on purpose.

Impact and usefulness

For the expert

Records the work **once, in their own words**, so the same explanation is never repeated.

For the new user

Steps come in their language and can be followed **hands-free**, even when the expert is away.

For the workplace

New people learn the process **without waiting for someone to explain it again**.

Wider impact

With **98%** of Indian internet users consuming Indian-language content, offline, local-language instructions help close the gap.

Beyond workshops

Kitchens, garages, labs, farms and machine setups: **anywhere the same explanation is repeated**.

Why it is feasible

- **We listen for steps instead of understanding the whole video.** People say “phir”, “ab” or “then” while explaining. Those words and the pauses split the take. No model to train.
- **The review screen makes it easier.** The expert fixes the cuts before signing off, so they only need to be close, not perfect.
- **The expert’s own voice.** Each step plays the original recording, in the language it was spoken.
- **No AI makes decisions.** Four modes are chosen by plain rules, so switching is fast and predictable.
- **Every model runs on the phone,** and any one can be missing. The app drops a tier and carries on.

Who is building

JAYAVARAPU SRI CHARAN

On-device AI, systems

Owns the pipeline, the models and the build.

JADALA SOHAN

UI and UX

Builds the screens the judge touches.

SURISETTI PREM SAI

Office Kit, content, demo, pitch

Owns the tuning file, the guides and the pitch.

Research

Learning from Narrated Instruction Videos (Alayrac et al.)

Narration and video together reveal a task's steps. → **We took the narration half and dropped the model.**

ARify (CHI 2026)

Turns narrated instructional videos into AR tutorials, but making them needs AR authoring expertise. → **The problem is current. They need AR; we need a phone.**

Internet in India 2024 (IAMAI-Kantar)

886M active users, 488M of them rural, and 98% access content in Indian languages. → **Instructions still arrive as English text.**

Thank you

TaskLens · Team Nexura

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